Statistical Foundations of Machine Learning

Practice Set -Joint distributions, Expectation, Variance, and Covariance

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1 Practice problems

Note: problems in this section are for practice, solutions not to be submitted

1. There are two jointly continuous random variables X and Y with the joint pdf-

$$f_{XY}(x,y) = \begin{cases} \frac{x^2}{4} + \frac{y^2}{4} + \frac{xy}{6} & 0 \le x \le 1, 0 \le y \le 2 \\ 0 & otherwise \end{cases}$$

   For $0 \le y \le 2$, find:

   (a) The conditional PDF of X given $Y = y$.

   (b) $P(X < \frac{1}{2} | Y = y)$

2. Two discrete random variables x and y have the given joint distribution.
   Find the following ($F_{xy}(x,y)$ represents the joint CDF) :

	x=1	x=2	x=3
y=1	3/12	1/12	3/12
y=2	1/12	3/12	1/12

   (a) $F_{xy}(1,1)$

   (b) $F_{xy}(1.5,4)$

   (c) $F_{xy}(4,9)$

   (d) $F_{xy}(1,-1.5)$

3. The random variables X, Y have joint PDF $f_{X,Y}(x,y)$:

$$f_{X,Y}(x,y) = \begin{cases} cx(y-x)e^{-y} & 0 \le x \le y \\ 0 & \text{elsewhere} \end{cases},$$

   where c is a constant.

   (a) Compute the value of c.

   (b) Obtain the conditional density function $f_{X|Y}(x|y)$.

4. Consider two independent random variables $X_1 \sim \text{Exp}(\lambda_1)$ and $X_2 \sim \text{Exp}(\lambda_2)$. Find the probability $\mathbb{P}(X_1 \le X_2)$.

   $X \sim \text{Exp}(\lambda)$: $f_X(x) = \lambda e^{-\lambda x}$ for $x \ge 0$

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5. Let N be a geometric random variable with PMF

$$p_N(k) = p(1-p)^{k-1}, k = 1, 2, \cdots, \ p \in (0, 1).$$

Find the conditional PMF $p_N(k|A) = \mathbb{P}\{N = k | N \in A\}$ if $A = \{N \text{ is even}\}$.

6. Suppose that n tubelights in a room are turned on at time 0. The lifetimes of the tubelights are independent and are exponentially distributed with parameter μ.

   (a) What is the cdf of the time after which the first tubelight fuses out?

   (b) Find the cdf of the time after which the room goes dark?

7. The number of pages X in a document has PMF

$$P_X(x) = \begin{cases} 0.15 & \text{if } x = 1, 2, 3, 4 \\ 0.1 & \text{if } x = 5, 6, 7, 8 \\ 0 & \text{otherwise} \end{cases}$$

A firm sends all documents with an even number of pages to printer A and all documents with an odd number of pages to printer B.

   (a) Find the conditional PMF of the length X of a document, given the document was sent to printer A.

   (b) Find the conditional PMF of the length X of a document, given the document was sent to printer B and had no more than six pages.

8. A positive integer is selected, with $P(I = n) = \frac{1}{2^n}, n = 1, 2, \cdots$. If I takes the value n, a coin with probability of heads equal to e^{-n} is tossed once. Find the probability that the resulting toss will be a head.

9. Let $f(x, y) = cx^2y$ for $x^2 \leq y \leq 1$ denote the joint pdf for (X, Y). Find:

   (a) c

   (b) $\mathbb{P}[|X| \geq Y]$

   (c) Marginals $f_X(x)$ and $f_Y(y)$

10. Let X be the random variable denoting the number of tosses of a biased coin to get the first head. The first head has not occurred in the first k tosses. What is the probability that you will have to toss the coin n more times.

11. Consider an undirected random graph $\mathcal{G}$, which is constructed as follows. $\mathcal{G}$ has n nodes and each of the $\binom{n}{2}$ possible edges is included in $\mathcal{G}$ with probability p, independent of all other edges. Define

$$I_i = \begin{cases} 1 & \text{if node } i \text{ is isolated i.e., there is no edge incident on node } i \\ 0 & \text{otherwise} \end{cases}$$

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(a) For $i \neq j$, are I_i, I_j independent? Provide mathematical argument for your answer.

(b) Find the expected number of isolated nodes in $\mathcal{G}$.

(c) Denote the number of edges in $\mathcal{G}$ by E. Find $\mathbb{E}[E]$.

12. If $\mathbb{E}[X] = 1$ and $\text{Var}(X) = 5$, find

(a) $\mathbb{E}[(2 + X)^2]$

(b) $\text{Var}(4 + 3X)$

13. We are given three independent random variables X,Y and Z. They follow the distributions $X \sim N(\mu, \sigma^2)$, and $Y, Z \sim \text{Uniform}(0, 2)$. We are given that -

- $E[X^2Y + XYZ] = 13$
- $E[XY^2 + ZX^2] = 14$

Find the numerical values of μ, σ.

14. **Balls and Bins** Consider k balls and n bins ($k \geq n$), and say we place each ball independently and uniformly at random in to one of the n bins. Denote the number of empty bins by B.

(a) What is the expected value of B?

(b) What is the variance of B?

15. Consider discrete random variables X, Y with the following joint pmf:

		Y		
		-1	0	1
X	0	$1/6$	$1/6$	0
	1	0	0	$1/3$
	3	$1/6$	$1/6$	0

(a) Find the covariance matrix for (X, Y).

(b) Are X, Y independent? Justify your answer

(c) Are X, Y uncorrelated? Justify your answer

(d) Are X, Y orthogonal? Justify your answer

16. Suppose that n light bulbs in a room are turned on at time 0. The lifetime of bulb are independent and exponentially distributed with parameter μ.

(a) What is the expected time after which the first bulb fuses out?

(b) What is the expected time after which the room goes dark?

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17. **(Mixed distributions)**
Let $X \sim \text{Bernoulli}(p)$, $Y \sim \mathcal{N}(0,1)$, and assume X and Y are independent.
Find $\text{Variance}((3X-1)Y)$.

18. **(Correlation – data science setting)**
In a simple linear regression, say a feature X is uniformly distributed on $[0,1]$. The target variable Y is generated as

$$Y = 2X + \epsilon,$$

where $\epsilon \sim \mathcal{N}(0,1)$ represents independent noise. Find the **correlation coefficient** between X and Y.

19. (**Dataset coverage under random sampling**)

A dataset $\mathcal{D}$ contains n distinct data points. A data scientist samples k data points independently **with replacement**, each selected uniformly at random from the dataset.

Let U denote the number of data points in the dataset $\mathcal{D}$ that are never sampled.

(a) Find $\mathbb{E}[U]$.

(b) Define $\hat{p} = U/n$ as an estimator for the fraction of data points not seen in the sample. Compute $\mathbb{E}[\hat{p}]$.

(c) Is $\hat{p}$ an unbiased estimator for the probability that a given data point is not sampled?

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